

12.684

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October 12, 2025

Question

If $\mathbf{A}$ is a square matrix then the orthogonality property mandates:

 $\mathbf{A}\mathbf{A}^T = \mathbf{I}$

 $\mathbf{A}\mathbf{A}^T = \mathbf{0}$

 $\mathbf{A}\mathbf{A}^T = \mathbf{A}^{-1}$

 $\mathbf{A}\mathbf{A}^T = \mathbf{A}$

A square matrix $\mathbf{A}$ is orthogonal if its transpose equals its inverse:

$$\mathbf{A}^T = \mathbf{A}^{-1} \quad (1)$$

Multiplying both sides on the left by $\mathbf{A}$:

$$\begin{aligned} \mathbf{A}\mathbf{A}^T &= \mathbf{A}\mathbf{A}^{-1} \\ \implies \mathbf{A}\mathbf{A}^T &= \mathbf{I} \end{aligned} \quad (2)$$

C Code

```
#include <stdio.h>
#include <math.h>

#define N 3 // Dimension of the square matrix
#define EPS 1e-6 // Tolerance for floating-point comparison

int main() {
    int i, j, k;
    double A[N][N] = {
        {1/sqrt(2), -1/sqrt(2), 0},
        {1/sqrt(2), 1/sqrt(2), 0},
        {0, 0, 1}
    };
    double AA_T[N][N] = {0};

    // Compute AA^T
    for(i = 0; i < N; i++) {
        for(j = 0; j < N; j++) {
```

C Code

```
        for(k = 0; k < N; k++) {
            AA_T[i][j] += A[i][k] * A[j][k];
        }
    }
}

int isOrthogonal = 1;
for(i = 0; i < N; i++) {
    for(j = 0; j < N; j++) {
        if(i == j && fabs(AA_T[i][j] - 1) > EPS) isOrthogonal
            = 0;
        if(i != j && fabs(AA_T[i][j]) > EPS) isOrthogonal = 0;
    }
}

if(isOrthogonal)printf(Matrix A is orthogonal);
else
printf(Matrix A is not orthogonal);
return 0;
}
```

```
import numpy as np

# Define a square matrix A
A = np.array([
    [1/np.sqrt(2), -1/np.sqrt(2), 0],
    [1/np.sqrt(2), 1/np.sqrt(2), 0],
    [0, 0, 1]
])

# Compute A * A^T
AA_T = A @ A.T

# Check if AA^T is identity
if np.allclose(AA_T, np.eye(A.shape[0])):
    print(Matrix A is orthogonal.)
else:
    print(Matrix A is not orthogonal.)
```